# Appendix

## Model selection and fit

Table 1: Model comparison results

| | Model 1 | | Model 2 | | Model 3 | | Model 4 | | Model 5 | |
|---|---|---|---|---|---|---|---|---|---|---|
| Term | Mean | SE | Mean | SE | Mean | SE | Mean | SE | Mean | SE |
| Barrier: Organic availability | | | | | 0.242 | 0.001 | 0.173 | 0.001 | 0.171 | 0.001 |
| Barrier: Undesirable traits | | | | | 0.105 | 0.001 | 0.124 | 0.001 | 0.121 | 0.001 |
| Barrier: Buyer requirements | | | | | 0.092 | 0.001 | 0.126 | 0.001 | 0.121 | 0.001 |
| Barrier: Insufficient quantity | | | | | 0.028 | 0.001 | 0.018 | 0.001 | 0.016 | 0.001 |
| Year: 2015 (vs. 2010) | -0.371 | 0.002 | | | | | -0.19 | 0.002 | -0.175 | 0.002 |
| Year: 2020 (vs. 2010) | -0.413 | 0.002 | | | | | -0.309 | 0.002 | -0.316 | 0.002 |
| Values organic seed | | | -0.27 | 0.001 | -0.202 | 0.001 | -0.202 | 0.001 | -0.200 | 0.001 |
| Farm size (acres) | | | 0.072 | 0.001 | 0.029 | 0.001 | 0.106 | 0.001 | 0.118 | 0.001 |
| Crop diversity | | | -0.035 | 0.001 | -0.064 | 0.001 | -0.071 | 0.001 | -0.068 | 0.001 |
| Seed saved or traded (%) | | | | | | | | | -0.095 | 0.001 |
| Certifiers request organic seed use | | | | | | | | | 0.038 | 0.001 |
| Forage crops (vs. Field) | 0.82 | 0.002 | | | | | 0.932 | 0.002 | 0.903 | 0.002 |
| Vegetable crops (vs. Field) | 0.186 | 0.002 | | | | | 0.286 | 0.002 | 0.248 | 0.002 |
| Western region (vs. North Central) | 0.439 | 0.002 | | | | | 0.3 | 0.002 | 0.338 | 0.002 |
| Northeastern region (vs. North Central) | 0.325 | 0.002 | | | | | 0.25 | 0.002 | 0.249 | 0.002 |
| Southern region (vs. North Central) | 0.252 | 0.003 | | | | | 0.116 | 0.003 | 0.122 | 0.003 |
| (Intercept) | -0.595 | 0.002 | -0.455 | 0.001 | -0.466 | 0.001 | -0.702 | 0.002 | -0.701 | 0.002 |

Table 2: Goodness of fit across models

| Model | WAIC | Maximum Likelihood | CPO |
|---|---|---|---|
| Model 1 | -4310 | 2114 | 1.23 |
| Model 2 | -4247 | 2100 | 1.21 |
| Model 3 | -4354 | 2130 | 1.24 |
| Model 4 | -4478 | 2157 | 1.28 |
| Model 5 | -4484 | 2148 | 1.28 |

## Full model results

Table 3: Full model results

| Term | Mean estimate | Standard error |
|---|---|---|
| Barrier: Organic availability | 0.171 | 0.001 |
| Barrier: Undesirable traits | 0.121 | 0.001 |
| Barrier: Buyer requirements | 0.121 | 0.001 |
| Barrier: Insufficient quantity | 0.016 | 0.001 |
| Year: 2015 (vs. 2010) | -0.175 | 0.002 |
| Year: 2020 (vs. 2010) | -0.316 | 0.002 |
| Values organic seed | -0.200 | 0.001 |
| Farm size (acres) | 0.118 | 0.001 |
| Crop diversity | -0.068 | 0.001 |
| Seed saved or traded (%) | -0.095 | 0.001 |
| Certifiers request organic seed use | 0.038 | 0.001 |
| Forage crops (vs. Field) | 0.903 | 0.002 |
| Vegetable crops (vs. Field) | 0.248 | 0.002 |
| Western region (vs. North Central) | 0.338 | 0.002 |
| Northeastern region (vs. North Central) | 0.249 | 0.002 |
| Southern region (vs. North Central) | 0.122 | 0.003 |
| (Intercept) | -0.701 | 0.002 |

## Predicted conventional seed use

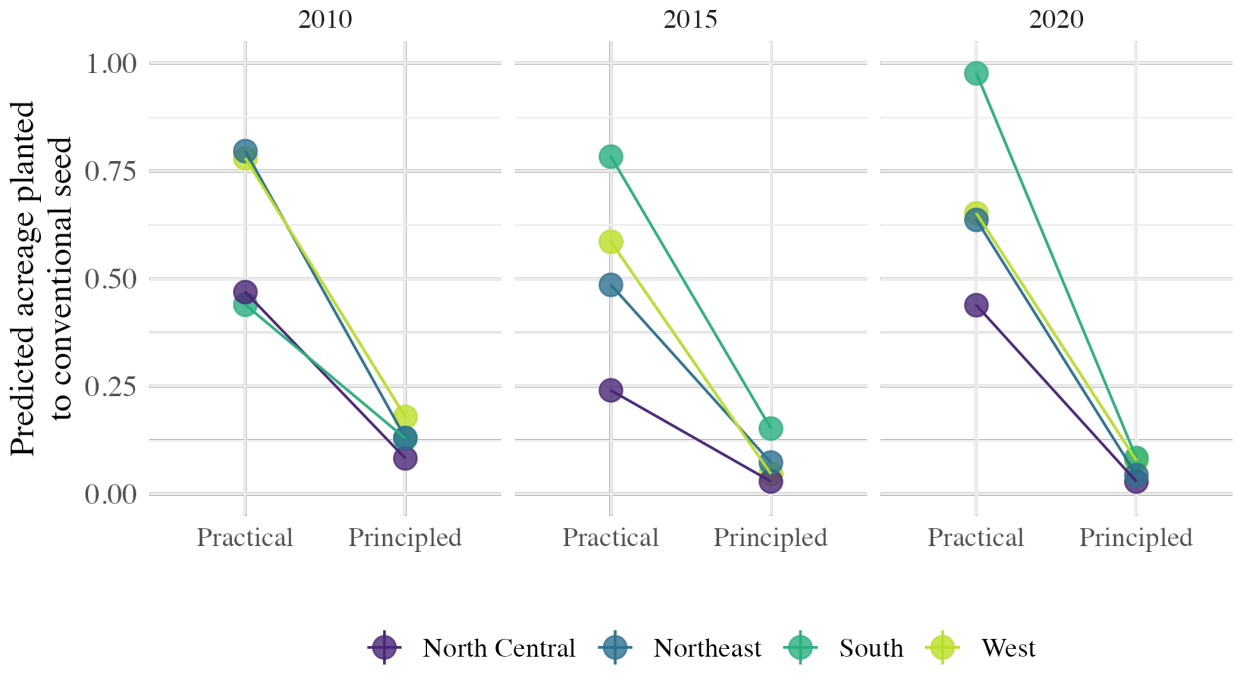



Figure 1: Predicted conventional seed use for field crop growers of two farmer profiles.

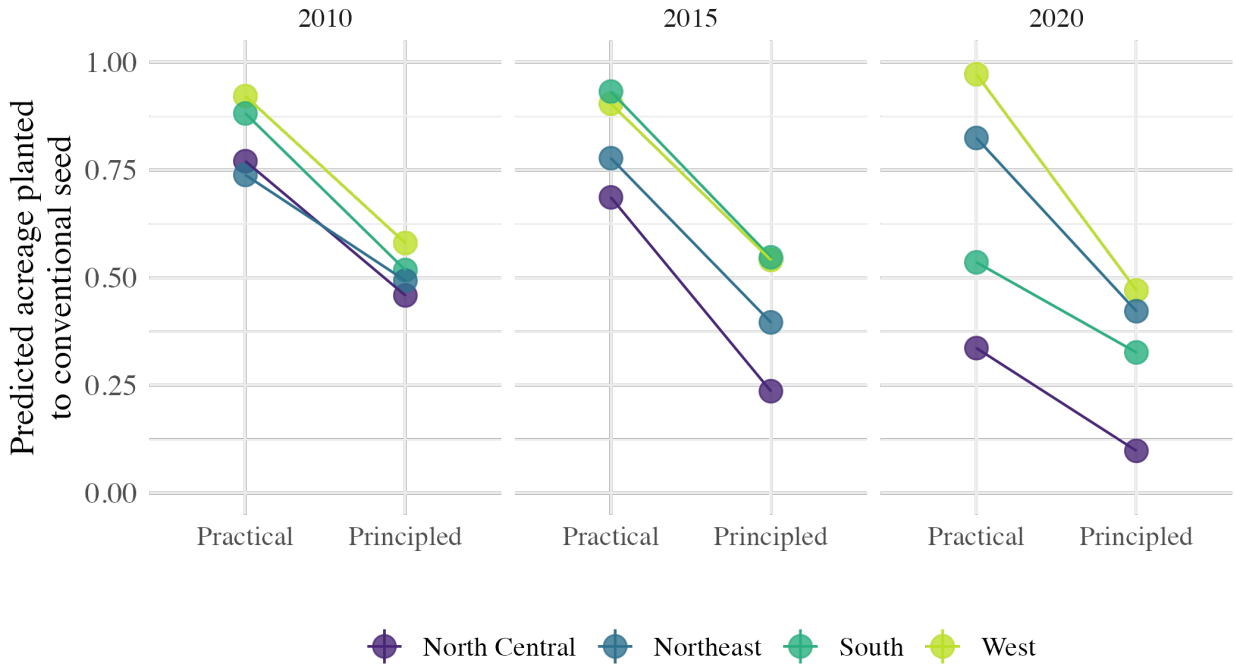



Figure 2: Predicted conventional seed use for forage crop growers of two farmer profiles.